

Research and teaching statement

Media systems are becoming spatial, multimodal and generative. That shifts the central question of media informatics: it is no longer enough to show that a system works — it has to be shown that people understand it, trust it appropriately and can use it safely. My work answers that question constructively and empirically at once: I build interactive media systems and develop the methods by which their effect can be measured reproducibly.

The thread runs through everything I have done. From the neural correlates of perceived speech quality, through Quality of Experience in networks and games, to today's XR, digital twin and agent systems, I have worked on the same question: how the subjective experience of technical media can be made objectifiable. That these methods went through ITU-T into international standardization (P.812, P.IntVR, G.1035) is the evidence that they carry beyond my own group.

THREE RESEARCH LINES

A — Spatial and multimodal interaction

How do people perceive, navigate and collaborate in mixed-reality environments?

WHAT IS DONE

Spatial audio in AR navigation (IEEE VR 2026); collaborative AR under spatial misalignment and latency (QoMEX 2024/2025); passthrough quality of mixed-reality devices; hand versus controller interaction; social acceptability of mobile XR in public; mixed reality at the workplace and in public space (IEEE ISMAR 2024).

WHAT COMES NEXT

Models and metrics for spatial coordination in shared mixed-reality spaces; interaction that adapts to attentional state and ambient conditions; and the transfer of laboratory findings to field conditions — a problem I most recently addressed under the title *Too Immersive for the Field?*, on safety risks in XR field studies.

B — Generative AI in media contexts

Generative methods produce media content, conversational partners and whole environments. The decisive question is not whether that succeeds, but under what conditions people can judge the result appropriately.

WHAT IS DONE

Photorealistic, AI-driven MetaHuman avatars for training situations (IEEE ISMAR 2024); emotion-adaptive virtual health assistants and their effect on social presence and trust; avatar design and size as factors in the user experience; GAN-based synthetic time-series data (Diversity and Societal Impact Award 2023); a scenario-based interview study on the explainability of medical AI.

WHAT COMES NEXT

Calibrated trust in generative media systems; design means that make the reliability and provenance of content perceivable rather than hiding them; multimodal detectability of synthetic media; and generative tools for authoring XR content, which addresses the largest bottleneck immersive applications currently have.

C — Evaluation methodology as the methodological core

This line is the core, and the one most useful to everyone else. It covers behavioural, psychophysiological and model-based methods: gaze, heart rate, EEG, interaction traces and subjective scales, brought together into defensible statements about quality, workload, presence, trust and safety.

WHAT IS DONE

Instruments developed and validated: pictographic scales for rapid affect capture in VR, questionnaires administered inside virtual environments, the Storytime dataset — and parts of this carried into ITU-T recommendations.

WHAT COMES NEXT

An open, documented repository of methods and data for the user-side assessment of media systems, usable by other groups; standards-conformant test procedures for generative and collaborative systems; and reproducibility in XR research, an unsolved problem in the field that my standardization experience is the right instrument for.

FUNDING STRATEGY

I arrive with running, transferable projects and a plan for the first five years.

Years 1–2 · Groundwork

Continuation of the running EU and state projects on adaptive XR safety training; a first DFG individual grant on evaluation methodology for generative and collaborative media systems; participation in local interdisciplinary centre proposals.

Years 2–4 · Broadening

Horizon Europe participation in Cluster 4 on the basis of the existing digital-twin consortia; a BMBF collaborative project on human-technology interaction; transfer projects with regional companies.

Years 3–5 · Consolidation

Preparation of a DFG research training group or research unit on trustworthy immersive systems, together with local partners; expansion of international consortium work.

Expanding the DFG side is a deliberate priority. My funding history so far is strongly European, ministry-shaped and application-oriented. At a university the basic-research line belongs alongside it; for the first two proposals the question and the preliminary work already exist.

HOW I TEACH

Media informatics is not learned by listening. Students have to design an interactive system, build it, test it on people, and then bear and explain the gap between what they expected and what they observed. That three-step — construct, measure, revise — is the core of my teaching. It is also why I do not hive empirical methods off into a separate methods module but put them inside the development projects.

— Foundations and specialisation must not stand side by side unconnected. Someone who learns in the first semester why an interface fails understands in the fifth why an architectural decision rescues it.

— Students should be able to question their own results. A user study that refutes your own hypothesis is the most valuable teaching experience of the whole degree.

— I teach the introductory courses gladly. They are not a burden but the point at which it is decided whether students can carry the specialisation later.

Project-based learning with real clients

Student projects run on genuine problems from companies, clinics and local government, drawn from my existing partner networks.

Research-led learning

At the Berlin University Alliance I led a student research group on user-centred tourist augmented reality across two semesters — an open question carried through to publication. I would offer that format again.

Immersive formats, with evidence

I use VR and AR in teaching and study their effect at the same time, among other things in language learning and in examination situations under VR conditions.

Generative AI, admitted and reflected

I explicitly admit AI tools in projects, but require students to document their use and to check the results.

Evaluating my own teaching

In 2026 I published a modular questionnaire for the audience-specific evaluation of course and transfer formats, and apply it to my own.

In supervision I work with clear milestones, documented expectations and regular, concrete feedback. Theses are carried through to publication where possible. For doctoral researchers: early first authorship, conference presence from the first year, a stay abroad with one of my international partners, and a realistic schedule. Supervising doctoral researchers is the most important task of a professorship.

INFRASTRUCTURE

I have built laboratory infrastructure for immersive research from scratch twice, most recently the Body Interaction Lab. A comparable laboratory would be modular: mixed-reality devices, motion and eye tracking, physiological measurement, a reconfigurable study room and a mobile unit for field studies with regional partners. I would run it expressly as shared infrastructure of the faculty, not as property of the chair.

The current version of this statement is at voigt-antons.de/research/statement/.